

Anubhav Das

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B.Tech. Mathematics & Computing, RGIPT | Graduating May 2027 | Open to relocation across India

EDUCATION

Rajiv Gandhi Institute of Petroleum Technology (Institute of National Importance)

Jais, Amethi

B.Tech. in Mathematics and Computing — CGPA: 7.29 / 10

2023 – 2027

- **Relevant coursework:** Data Structures & Algorithms, Design & Analysis of Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Discrete Mathematics, Linear Algebra, Probability & Statistics.

EXPERIENCE

India Meteorological Department (IMD), Ministry of Earth Sciences

New Delhi, India

Research Intern — Numerical Weather Prediction & Deep Learning

May – July 2025

- Researched and prototyped hybrid deep-learning architectures as candidate approaches for IMD's operational NWP problems — thunderstorm, sandstorm, and cyclone detection — grounding design choices in recent literature on spatiotemporal graph networks and attention-based weather models.
- Explored a U-Net + SWIN Transformer variant for thunderstorm prediction and a triple-convolution modification to the U-Net encoder block; discontinued the SWIN variant after it underperformed simpler alternatives on qualitative evaluation.
- Studied IMD's operational NCUM model for sandstorm forecasting to identify input variables unavailable in existing pipelines, and worked on deriving proxy inputs for them.
- Designed an MTGNN-U-Net hybrid as a proof-of-concept for sandstorm and cyclone detection: a graph-neural bottleneck learning directed edges between 256 spatial nodes (16×16 grid) over 7-day input sequences via dilated temporal inception layers and mix-hop graph convolutions; validated qualitatively against IMD's available observational data.

PROJECTS

Support AI — Multi-Tenant SaaS Customer Support Platform

Live | GitHub

Python, FastAPI, PostgreSQL, pgvector, Redis, RabbitMQ, Celery, Docker, Prometheus, Grafana

- Built and deployed a production multi-tenant backend exposing **73 REST operations** across 55 paths, with row-level tenant isolation, JWT authentication, and role-scoped access control.
- **Profiled and fixed a p95 latency regression:** load-tested at 50 concurrent connections (15,782 requests, 100% 2xx) for a 137 ms p95 at 567 req/s, then traced a degradation to 310 ms under PDF-upload load to synchronous parsing and chunking blocking the FastAPI event loop (164 ms per 4 MB file); moving the work off-loop restored p95 to **151 ms**.
- Implemented per-tenant RAG retrieval over **pgvector** — document upload, PDF parsing, semantic chunking, and cosine-similarity search scoped to per-tenant knowledge bases — with Celery/RabbitMQ workers handling webhook dispatch, transactional email, and scheduled cleanup.
- Containerised the stack with Docker Compose; 30+ Alembic migrations, structured logging via structlog, Prometheus/Grafana instrumentation, and 57 pytest-asyncio unit tests.

oxideGate — OpenAI-Compatible LLM Inference Gateway

GitHub

Rust, Tokio, Axum, reqwest (in progress — Aug 2026 – present)

- Built a streaming HTTP reverse proxy for LLM backends with SSE passthrough and structured error mapping; sustains 15,000 req/s at p99 25 ms (100 concurrent connections) with no measurable latency added over a direct backend call, benchmarked with hey over 30s runs against a mock backend.
- Measured system behaviour under saturation against a capacity-limited backend to quantify the cost of having no admission control: raising offered concurrency 4→64 multiplied p50 latency 16× (608 ms → 9.7 s) while throughput stayed flat at 6.6 req/s and zero requests were shed — the measurement motivating a bounded-queue scheduler, currently designed but not yet implemented.
- 7 integration tests covering streaming, error mapping, and request validation against a live stub backend; cargo clippy -D warnings and cargo fmt clean.

Weather Forecasting U-Net — Tornado Probability Mapping

GitHub

Python, PyTorch, NOAA Reanalysis Data, NetCDF, xarray

- Built a 4-level U-Net (64→128→256→512→1024 channels) with skip connections mapping CAPE, CIN, and tropopause geopotential height to tornado and significant-tornado probability as 256×256 spatial heatmaps, trained on a full year of NOAA reanalysis data with a weighted BCE loss (pos_weight 50:100) addressing a ~1:228 pixel-level class imbalance (≥2% SPC threshold).
- Replaced an initial random train/val split with a chronological split after identifying that random splitting breaks temporal independence across autocorrelated daily weather fields; hardened five-dataset NetCDF ingestion against time-axis decoding mismatches.
- Root-caused a cross-module unit-scale bug: percent-valued labels were silently clamped into a near-binary [0,1] mask before reaching the loss function, corrupting an early training run without raising any error.
- Identified a geographic misregistration between the global input grid and the CONUS-only label grid from independent resizing, and documented the fix (common-grid regridding) required before any performance claim is meaningful.

TECHNICAL SKILLS

Languages: C++, Python, C, Rust, JavaScript, SQL

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Design, Operating Systems, DBMS, Computer Networks, Complexity Analysis

Backend & Systems: FastAPI, REST API design, PostgreSQL, Redis, RabbitMQ, Celery, async I/O, Docker, Git, Linux

ML & Data: PyTorch, NumPy, pandas, xarray, vector search (pgvector), Prometheus/Grafana

LEADERSHIP & ACHIEVEMENTS

- **Head, Entrepreneurship Cell, RGIPT** (Aug 2025 – present) — lead a 96-member team across 8 events; secured partnerships with 4 startups.
- **PR & Publicity Coordinator, URJA SANGAM** — drove outreach reaching 3M+ online views and 30+ colleges.
- **Finalist, IIT Bombay National Entrepreneurship Challenge** — top 50 of 500+ colleges nationally.